MILITARY SPECIFICATION

PENTAERYTHRITOL TETRANITRATE TABLETS, MODIFIED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Pentaerythritol Tetranitrate Tablets, Modified (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (see 6.3).

STANDARDS

Federal

FED-STD-140	Tablets (for Medicinal Purposes).

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

FSC 6505

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Pentaerythritol Tetranitrate Tablets, Modified, sustained type, and shall be in accordance with all applicable requirements of FED-STD-140 and as specified herein:

S2. Classification. Shall be type XIII, class 1.

S5.3 The following is added to this paragraph:

Shall be sustained-action dosage tablets containing in each tablet, within the designated assay limits for the tablets, 80 mg of Pentaerythritol Tetranitrate ($C_5H_8N_4O_{12}$), so prepared as to provide a therapeutic response for sustained periods of 8 to 12 hours.

S6.4.9.1 Disintegration shall comply with the release rate.

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3.1.1 Assay. The tablets shall assay to contain not less than 95.0 percent and not more than 105.0 percent of the required total amount of Pentaerythritol Tetranitrate when assayed as specified in 4.3.1.

3.1.2 Release rate. The tablets shall release an initial 1 hour dose of 17.0 to 23.0 mg of active medicament and a total release of 64.0 to 75.0 mg of active medicament in the first 5 continuous hours when determined as specified in 4.3.2.

3.1.3 Identification. Shall comply with the identification requirement of the N.F. for Pentaerythritol Tetranitrate Tablets.

3.2 Ingredients.

3.2.1 Pentaerythritol tetranitrate. The pentaerythritol tetranitrate used in the manufacture of the tablets shall comply with the requirements of the N.F. for diluted pentaerythritol tetranitrate.

3.2.2 Other ingredients. All other ingredients entering into the finished tablets shall comply with paragraph S5.1 of FED-STD-140.

3.3 Expiration dating period. The expiration dating period shall be 60 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

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4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. The sampling as shown in FED-STD-140 shall be applicable.

4.3 Tests.

4.3.1 Assay for Pentaerythritol Tetranitrate.

Reagents:

Potassium Nitrate Standard Solution. Dissolve 127.9 mg of dried potassium nitrate in approximately 3 ml of purified water. Dilute with glacial acetic acid to exactly 200 ml. One (1) ml is equivalent to 0.5 mg of Pentraerythritol Tetranitrate.

Phenoldisulfonic Acid Reagent. Phenoldisulfonic Acid Purified (Fisher), or Harleco Reagent shall be used. If the commercial product is not available, prepare the reagent as follows:

Dissolve 50 grams of phenol crystals in 300 ml of concentrated sulfuric acid; add 150 ml of fuming sulfuric acid (20 to 23 percent SO_3), and heat to 100°C. Maintain the mixture at this temperature for 2 hours, cool, and store in glass-stoppered bottles.

Glacial Acetic Acid, U.S.P.

Ammonium Hydroxide 10 percent solution. Dilute 400 ml of ammonium hydroxide stronger solution to 1000 ml with purified water.

Procedure:

Place 6 tablets into a 250-ml beaker, add approximately 100 ml of glacial acetic acid and heat on a steam bath for approximately 1 hour, using a glass stirring rod to effect complete disintegration. Transfer the turbid solution, quantitatively, to a dry 200-ml volumetric flask, rinsing the beaker with glacial acetic acid. Cool to room temperature, dilute to exactly 200 ml with glacial acetic acid, and mix thoroughly. Pipet a 10-ml aliquot into a dry 50-ml volumetric flask, dilute to exactly 50 ml with glacial acetic acid and mix thoroughly.

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Pipet 1 ml of this sample solution into a 100-ml volumetric flask; 1 ml of glacial acetic acid into a second 100-ml volumetric flask; and add 1 ml of standard KNO_3 solution into a third 100-ml volumetric flask. Add 2 ml of Phenoldisulfonic Acid Reagent to each flask, mix, and allow to stand for 5 minutes at room temperature. Add approximately 25 ml of purified water and 20 ml of NH_4OH (10%) to each flask, cool, dilute to 100 ml with purified water and mix thoroughly. Filter approximately 50 ml of sample solution (standard and blank are not filtered) through Whatman No. 40 filter paper, discarding the first 25 ml of filtrate and determine the absorbances of the sample and standard solutions in 1-cm cells at a maximum of about 409 nm with a suitable spectrophotometer, using a reagent blank.

Calculation:

$$\frac{Au}{As} \times 0.5 \times \frac{50}{10} \times \frac{200}{6} = \text{mg Pentaerythritol Tetranitrate per tablet}$$

Where: Au is absorbance of sample
As is absorbance of the standard

4.3.2 Release Rate.

4.3.2.1 Assay for Pentaerythritol Tetranitrate released during 1 hour.

Place ten tablets into each of four 90-ml capacity, capped, bottles containing 80 ml of Simulated Gastric Fluid, T.S. Rotate the capped bottles in a constant temperature bath at 37° C. ± 2°C. for one (1) hour, by means of the rotating bottle apparatus (see Drug Standard, Vol. 26, No. 3, pages 79-80). At the end of one hour, remove the fluid contents from each of the bottles, and wash the residues from each bottle with 25 ml of purified water. Return the washed residues from two of the bottles into separate 90-ml bottles, and retain for assay for Pentaerythritol Tetranitrate released during 5 hours. Transfer the remaining residues to a dry beaker, and allow to air-dry overnight. Transfer quantitatively, 6 of the dry residues from one tube to a dry 250-ml beaker. Add approximately 100 ml of glacial acetic acid and heat on a steam bath for about 1 hour, using a glass stirring rod to effect

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complete disintegration. Transfer, quantitatively, the turbid solution to a dry 200-ml volumetric flask rinsing the beaker with glacial acetic acid. Cool to room temperature, dilute to exactly 200 ml with glacial acetic acid and mix thoroughly. Pipet a 15 ml aliquot into a dry 50-ml volumetric flask, dilute to exactly 50 ml with glacial acetic acid and mix thoroughly. Proceed as directed under assay for total Pentaerythritol Tetranitrate beginning with "Pipet 1 ml of this sample solution," etc.

Calculation:

$$\text{Total Assay} - \frac{Au}{As} \times 0.5 \times \frac{50}{15} \times \frac{200}{6} = \text{mg Pentaerythritol Tentranitrate per tablet released in 1 hour.}$$

Where Au is abosrbance of sample
As is absorbance of standard
Total assay is the total mg of Pentaerythritol Tetranitrate per tablet

4.3.2.2 Assay for Pentaerythritol Tetranitrate released during 5 hours.

To each of the two 90-ml bottles, with residues retained from test for Pentaerythritol Tetranitrate, released during 1 hour, add 80 ml of Simulated Intestinal Fluid, T.S. Rotate the capped bottles in a constant temperature bath at 37° ± 2°C. for 2 hours. At the end of this period, remove 20 ml of the fluid contents from each bottle, recap, and continue rotation in the 37° ± 2°C. bath for two (2) additional hours. At the end of this period (5 hours total), remove the fluid contents from each of the bottles, wash the residues from each bottle with 25 ml of purified water, transfer the residues to a dry beaker, and allow to dry overnight. Utilizing the residues from 1 tube, add approximately 75 ml of glacial acetic acid and heat on a steam bath for 1 hour, using a glass stirring rod to effect disintegration. Transfer, quantitatively, the turbid solution to a 100-ml volumetric flask, rinsing the beaker with glacial acetic acid. Cool to room temperature, dilute to exactly 100 ml with glacial acetic acid, and mix thoroughly. Pipet a 25-ml aliquot into a dry 50-ml volumetric flask, dilute to exactly 50 ml with glacial acetic acid, and mix thoroughly. Proceed as directed under assay for total Pentaerythritol Tetranitrate, beginning with "Pipet 1 ml of this sample solution," etc.

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Calculation:

$$\text{Total assay} - \frac{Au}{As} \times 0.5 \times \frac{50}{25} \times \frac{100}{10} = \text{Pentaerythritol Tetranitrate per tablet released in 5 hours}$$

Where:

Au is absorbance of sample
As is absorbance of standard
Total assay is the total mg of Pentaerythritol Tentranitrate per tablet

4.4 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (see 6.3) and as specified herein.

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5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing one thousand (1000) tablets as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate containers. Immediate containers shall be a commercially available immediate container. Containers shall be considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Four (4) units of issue shall be packaged as specified in 5.2.9 of PPP-C-186.

5.2.3 Level C. Unit shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Twenty-four (24) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twenty-four (24) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carrier to the final destination called for in the contract or purchase order.

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5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

5.4.1.1 Immediate container. The label of the immediate container shall bear the following:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package. Each unit package shall bear the following:

(a) labeling information in accordance with commercial practice.

(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with 5.5.3 and 5.5.8.1 of PPP-C-186.

5.4.3 Exterior container. Exterior container shall be marked in accordance with 5.5.4 and 5.5.8.1 of PPP-C-186.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

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6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-404-2658	PENTAERYTHRITOL TETRANITRATE TABLETS, MODIFIED, 80 mg, 1000s

6.3 All references to PPP-C-186 refer to Interim Federal Specifications PPP-C-00186B which is mandatory for use until superseded by a revision.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1461

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